Lab Assignment 1: Basic Statistics & Visualization in R

Objective: Learn how to calculate descriptive statistics and generate foundational graphics in R.
Dataset: Built-in R dataset `mtcars` (32 cars, 11 variables).
Tasks: (1) Summary statistics - Mean, Median, Standard Deviation, Variance, IQR; (2) Histogram; (3) Boxplot to identify outliers across groups; (4) Scatter plot with trend line.
Run with: `Rscript lab1_basic_stats_visualization.R`

1. R Code

```r
# ============================================================================
# Lab Assignment 1: Basic Statistics & Visualization in R
# ----------------------------------------------------------------------------
# Objective : Calculate descriptive statistics and generate
#             foundational graphics in R.
# Dataset   : Built-in R dataset `mtcars`
# Run with  : Rscript lab1_basic_stats_visualization.R
#             (or source() it inside RStudio)
# Output    : Step-by-step console log + 3 PNG plots in ./plots/
# ============================================================================

# ---------- Helper: print a clearly visible step header --------------
step <- function(n, title) {
  cat("\n", strrep("=", 70), "\n", sep = "")
  cat(sprintf("STEP %s : %s\n", n, title))
  cat(strrep("=", 70), "\n", sep = "")
}

# Save each plot as PNG; when running interactively also draw on screen
out_dir <- "plots"
if (!dir.exists(out_dir)) dir.create(out_dir)

save_plot <- function(file, draw) {
  path <- file.path(out_dir, file)
  png(path, width = 900, height = 650, res = 110)
  draw()
  dev.off()
  if (interactive()) draw()
  cat(">> Plot saved to:", path, "\n")
}

cat("Lab Assignment 1 - Basic Statistics & Visualization in R\n")
cat("R version :", R.version.string, "\n")
cat("Run on    :", format(Sys.time(), "%Y-%m-%d %H:%M:%S"), "\n")


# ============================================================================
# STEP 1 : Load the dataset
# ============================================================================
step(1, "Load the built-in 'mtcars' dataset")

data(mtcars)
cat(">> Dataset loaded successfully.\n")
cat(">> Dimensions (rows x columns):", nrow(mtcars), "x", ncol(mtcars), "\n\n")

cat(">> Structure of the dataset:\n")
str(mtcars)

cat("\n>> First 6 rows:\n")
print(head(mtcars))

cat("\n>> Missing values per column:\n")
print(colSums(is.na(mtcars)))


# ============================================================================
# STEP 2 : Summary statistics for key numerical attributes
# ============================================================================
step(2, "Summary statistics (Mean, Median, SD, Variance, IQR)")

key_vars <- c("mpg", "hp", "wt", "disp", "qsec")
cat(">> Key numerical attributes selected:", paste(key_vars, collapse = ", "), "\n")
```

```r
cat("   mpg  = Miles per gallon\n")
cat("   hp   = Gross horsepower\n")
cat("   wt   = Weight (1000 lbs)\n")
cat("   disp = Displacement (cu. in.)\n")
cat("   qsec = 1/4 mile time (seconds)\n")

cat("\n>> 2a. Built-in summary() output:\n")
print(summary(mtcars[, key_vars]))

cat("\n>> 2b. Calculating each statistic individually (example: mpg):\n")
cat("   mean(mtcars$mpg)   =", round(mean(mtcars$mpg), 3), "\n")
cat("   median(mtcars$mpg) =", round(median(mtcars$mpg), 3), "\n")
cat("   sd(mtcars$mpg)     =", round(sd(mtcars$mpg), 3), "\n")
cat("   var(mtcars$mpg)    =", round(var(mtcars$mpg), 3), "\n")
cat("   IQR(mtcars$mpg)    =", round(IQR(mtcars$mpg), 3), "\n")

cat("\n>> 2c. Descriptive statistics table for all key attributes:\n")
stats_table <- data.frame(
  Mean     = sapply(mtcars[, key_vars], mean),
  Median   = sapply(mtcars[, key_vars], median),
  Std_Dev  = sapply(mtcars[, key_vars], sd),
  Variance = sapply(mtcars[, key_vars], var),
  IQR      = sapply(mtcars[, key_vars], IQR),
  Min      = sapply(mtcars[, key_vars], min),
  Max      = sapply(mtcars[, key_vars], max)
)
print(round(stats_table, 3))

write.csv(round(stats_table, 3), "summary_statistics.csv")
cat("\n>> Statistics table saved to: summary_statistics.csv\n")


# ========================================================================
# STEP 3 : Histogram - distribution of a single continuous variable
# ========================================================================
step(3, "Histogram of Miles Per Gallon (mpg)")

h <- hist(mtcars$mpg, breaks = 8, plot = FALSE)
cat(">> Histogram bins (class intervals) and frequencies:\n")
print(data.frame(
  Bin       = paste0("(", head(h$breaks, -1), ", ", h$breaks[-1], "]"),
  Frequency = h$counts
))

save_plot("1_histogram_mpg.png", function() {
  hist(mtcars$mpg,
       breaks = 8,
       col    = "skyblue",
       border = "white",
       main   = "Histogram of Miles Per Gallon (mtcars)",
       xlab   = "Miles Per Gallon (mpg)",
       ylab   = "Frequency")
  abline(v = mean(mtcars$mpg),   col = "red",       lwd = 2, lty = 2)
  abline(v = median(mtcars$mpg), col = "darkgreen", lwd = 2, lty = 3)
  legend("topright",
         legend = c(sprintf("Mean = %.2f",   mean(mtcars$mpg)),
                    sprintf("Median = %.2f", median(mtcars$mpg))),
         col = c("red", "darkgreen"), lty = c(2, 3), lwd = 2, bty = "n")
})

cat(">> Interpretation: mean (", round(mean(mtcars$mpg), 2),
    ") > median (", median(mtcars$mpg),
    "), so the mpg distribution is slightly right-skewed.\n", sep = "")


# ========================================================================
# STEP 4 : Boxplot - identify potential outliers across groups
# ========================================================================
step(4, "Boxplot of Horsepower (hp) grouped by Number of Cylinders (cyl)")

cat(">> Number of cars in each cylinder group:\n")
print(table(cyl = mtcars$cyl))

cat("\n>> Five-number summary of hp for each cylinder group:\n")
print(tapply(mtcars$hp, mtcars$cyl, fivenum))

```

```r
b <- boxplot(hp ~ cyl, data = mtcars, plot = FALSE)

cat("\n>> Outliers detected (values beyond 1.5 * IQR from the box):\n")
if (length(b$out) > 0) {
  out_cyl  <- as.numeric(b$names[b$group])
  out_rows <- mtcars[paste(mtcars$cyl, mtcars$hp) %in% paste(out_cyl, b$out), ]
  print(out_rows[, c("cyl", "hp", "mpg", "wt")])
} else {
  cat("   No outliers found.\n")
}

save_plot("2_boxplot_hp_by_cyl.png", function() {
  boxplot(hp ~ cyl, data = mtcars,
          col  = c("lightgreen", "khaki", "salmon"),
          main = "Horsepower by Number of Cylinders (mtcars)",
          xlab = "Number of Cylinders",
          ylab = "Gross Horsepower (hp)",
          outpch = 19, outcol = "red", outcex = 1.3)
  if (length(b$out) > 0) {
    text(b$group, b$out, labels = rownames(out_rows),
         pos = 4, cex = 0.8, col = "red")
  }
})


# ======================================================================
# STEP 5 : Scatter plot with trend line
# ======================================================================
step(5, "Scatter Plot of Weight (wt) vs Miles Per Gallon (mpg) + Trend Line")

r <- cor(mtcars$wt, mtcars$mpg)
cat(">> Pearson correlation between wt and mpg: r =", round(r, 4), "\n")

model <- lm(mpg ~ wt, data = mtcars)
cat("\n>> Fitting linear regression model: lm(mpg ~ wt)\n")
print(summary(model))

coefs <- coef(model)
cat(sprintf(">> Trend line equation: mpg = %.3f + (%.3f) * wt\n", coefs[1], coefs[2]))
cat(sprintf(">> R-squared = %.4f\n", summary(model)$r.squared))

save_plot("3_scatter_wt_vs_mpg.png", function() {
  plot(mtcars$wt, mtcars$mpg,
       pch  = 19,
       col  = "steelblue",
       main = "Car Weight vs Fuel Efficiency (mtcars)",
       xlab = "Weight (1000 lbs)",
       ylab = "Miles Per Gallon (mpg)")
  abline(model, col = "red", lwd = 2)
  legend("topright",
         legend = c("Observed cars",
                    sprintf("Trend: mpg = %.2f %.2f*wt", coefs[1], coefs[2]),
                    sprintf("r = %.3f,  R^2 = %.3f", r, summary(model)$r.squared)),
         col = c("steelblue", "red", NA), pch = c(19, NA, NA),
         lty = c(NA, 1, NA), lwd = 2, bty = "n")
})

cat(">> Interpretation: strong negative relationship - heavier cars give",
    "lower mpg.\n   Each extra 1000 lbs reduces mpg by about",
    round(abs(coefs[2]), 2), "miles per gallon.\n")


# ======================================================================
# Done
# ======================================================================
step("6", "Lab Assignment 1 completed")
cat("Files generated:\n")
cat("  - summary_statistics.csv\n")
cat(paste0("  - ", file.path(out_dir, list.files(out_dir))), sep = "\n")
```

2. Execution Output (console printout)

```
Lab Assignment 1 - Basic Statistics & Visualization in R
R version : R version 4.3.3 (2024-02-29)
Run on    : 2026-10-26 21:07:32

======================================================================
STEP 1 : Load the built-in 'mtcars' dataset
======================================================================
>> Dataset loaded successfully.
>> Dimensions (rows x columns): 32 x 11

>> Structure of the dataset:
'data.frame':   32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : num  0 0 1 1 0 1 0 1 1 1 ...
 $ am  : num  1 1 1 0 0 0 0 0 0 0 ...
 $ gear: num  4 4 4 3 3 3 3 4 4 4 ...
 $ carb: num  4 4 1 1 2 1 4 2 2 4 ...
>> First 6 rows:
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
>> Missing values per column:
 mpg  cyl disp   hp drat   wt qsec   vs   am gear carb
   0    0    0    0    0    0    0    0    0    0    0

======================================================================
STEP 2 : Summary statistics (Mean, Median, SD, Variance, IQR)
======================================================================
>> Key numerical attributes selected: mpg, hp, wt, disp, qsec
   mpg  = Miles per gallon
   hp   = Gross horsepower
   wt   = Weight (1000 lbs)
   disp = Displacement (cu. in.)
   qsec = 1/4 mile time (seconds)

>> 2a. Built-in summary() output:
      mpg              hp              wt             disp
 Min.   :10.40   Min.   : 52.0   Min.   :1.513   Min.   : 71.1
 1st Qu.:15.43   1st Qu.: 96.5   1st Qu.:2.581   1st Qu.:120.8
 Median :19.20   Median :123.0   Median :3.325   Median :196.3
 Mean   :20.09   Mean   :146.7   Mean   :3.217   Mean   :230.7
 3rd Qu.:22.80   3rd Qu.:180.0   3rd Qu.:3.610   3rd Qu.:326.0
 Max.   :33.90   Max.   :335.0   Max.   :5.424   Max.   :472.0
      qsec
 Min.   :14.50
 1st Qu.:16.89
 Median :17.71
 Mean   :17.85
 3rd Qu.:18.90
 Max.   :22.90

>> 2b. Calculating each statistic individually (example: mpg):
   mean(mtcars$mpg)   = 20.091
   median(mtcars$mpg) = 19.2
   sd(mtcars$mpg)     = 6.027
   var(mtcars$mpg)    = 36.324
   IQR(mtcars$mpg)    = 7.375

>> 2c. Descriptive statistics table for all key attributes:
        Mean  Median Std_Dev  Variance     IQR    Min     Max
mpg   20.091  19.200   6.027    36.324   7.375 10.400  33.900
hp   146.688 123.000  68.563  4700.867  83.500 52.000 335.000
wt     3.217   3.325   0.978     0.957   1.029  1.513   5.424
disp 230.722 196.300 123.939 15360.800 205.175 71.100 472.000
```

```
qsec  17.849  17.710   1.787     3.193   2.008 14.500  22.900

>> Statistics table saved to: summary_statistics.csv
======================================================================
STEP 3 : Histogram of Miles Per Gallon (mpg)
======================================================================
>> Histogram bins (class intervals) and frequencies:
       Bin Frequency
1 (10, 15]         6
2 (15, 20]        12
3 (20, 25]         8
4 (25, 30]         2
5 (30, 35]         4
>> Plot saved to: plots/1_histogram_mpg.png
>> Interpretation: mean (20.09) > median (19.2), so the mpg distribution is slightly right-skewed.

======================================================================
STEP 4 : Boxplot of Horsepower (hp) grouped by Number of Cylinders (cyl)
======================================================================
>> Number of cars in each cylinder group:
cyl
 4  6  8
11  7 14

>> Five-number summary of hp for each cylinder group:
$`4`
[1]  52.0  65.5  91.0  96.0 113.0
$`6`
[1] 105 110 110 123 175

$`8`
[1] 150.0 175.0 192.5 245.0 335.0
>> Outliers detected (values beyond 1.5 * IQR from the box):
             cyl  hp  mpg   wt
Ferrari Dino   6 175 19.7 2.77
>> Plot saved to: plots/2_boxplot_hp_by_cyl.png

======================================================================
STEP 5 : Scatter Plot of Weight (wt) vs Miles Per Gallon (mpg) + Trend Line
======================================================================
>> Pearson correlation between wt and mpg: r = -0.8677

>> Fitting linear regression model: lm(mpg ~ wt)

Call:
lm(formula = mpg ~ wt, data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max
-4.5432 -2.3647 -0.1252  1.4096  6.8727

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.2851     1.8776  19.858  < 2e-16 ***
wt           -5.3445     0.5591  -9.559 1.29e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.046 on 30 degrees of freedom
Multiple R-squared:  0.7528,    Adjusted R-squared:  0.7446
F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10

>> Trend line equation: mpg = 37.285 + (-5.344) * wt
>> R-squared = 0.7528
>> Plot saved to: plots/3_scatter_wt_vs_mpg.png
>> Interpretation: strong negative relationship - heavier cars give lower mpg.
   Each extra 1000 lbs reduces mpg by about 5.34 miles per gallon.

======================================================================
STEP 6 : Lab Assignment 1 completed
======================================================================
Files generated:
  - summary_statistics.csv
  - plots/1_histogram_mpg.png
  - plots/2_boxplot_hp_by_cyl.png
  - plots/3_scatter_wt_vs_mpg.png
```

3. Output Plots

Figure 1: Histogram of Miles Per Gallon (mpg)

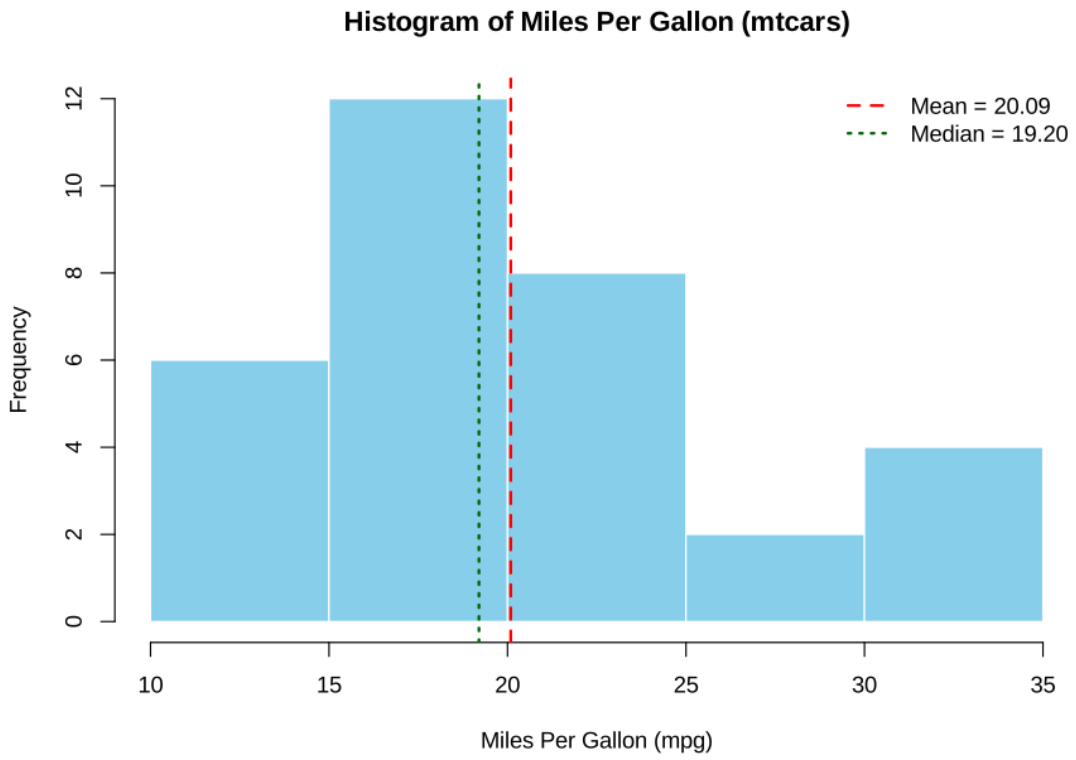


Figure 2: Boxplot of Horsepower by Number of Cylinders

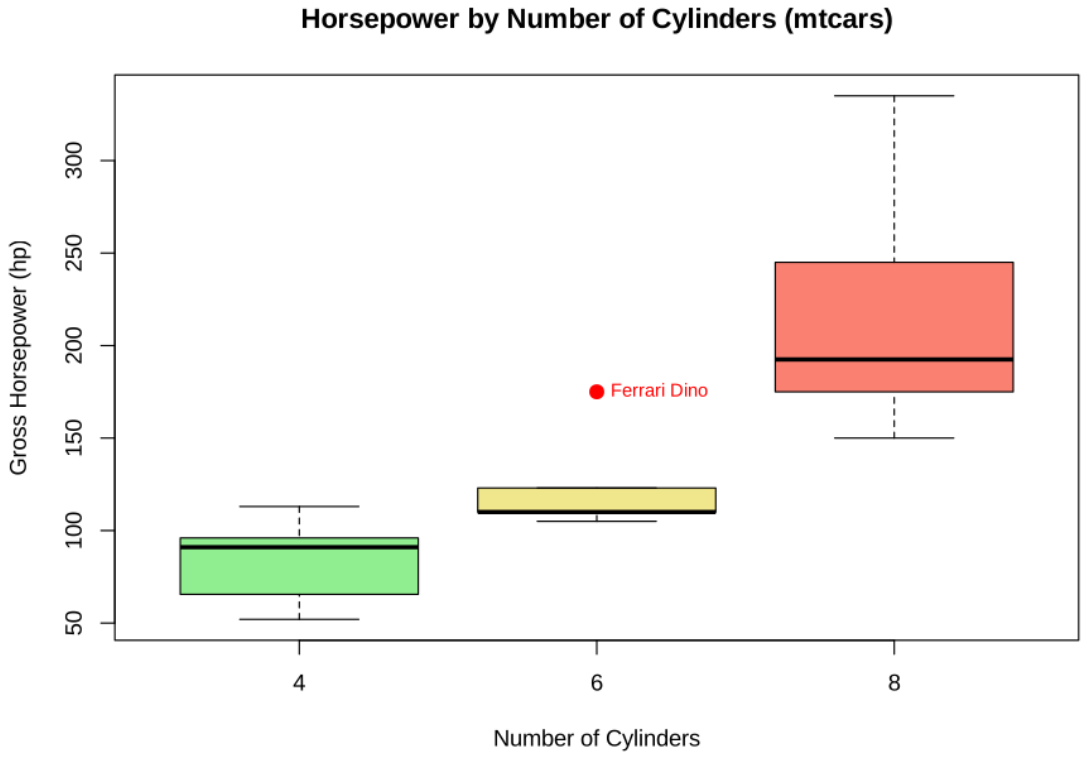


Figure 3: Scatter Plot of Weight vs mpg with Trend Line

